Elektroniske enheter og kretser
BJT Common-Emitter and Class-A Power Amplifiers

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***Abstract*—This report presents an investigation of BJT common-emitter and class-A power amplifiers. The work is divided into two parts. In the first part, the DC and AC behaviour of a common-emitter amplifier is studied, including bias voltages, voltage gain, input impedance, and output impedance, with measured results compared against calculated values. In the second part, the circuit is reconfigured as a class-A power amplifier, and its input power, output power, and efficiency are evaluated under maximum undistorted output conditions. The measurements were generally in good agreement with theoretical calculations, with most differences within ten per cent for bias and impedance parameters. For the power and efficiency analysis, larger discrepancies were observed, mainly due to transistor non-idealities and practical measurement limitations. The study demonstrates how theoretical analysis and experimental results can be combined to assess the performance and limitations of amplifier circuits.**

***Index Terms*—BJT amplifier, common-emitter, class-A, voltage gain, impedance, efficiency**

I. Introduction

The BJT common-emitter amplifier is a common transistor circuit, known for providing voltage gain with phase inversion and moderate input and output impedances. A clear understanding of its DC bias point and AC characteristics is important for predicting circuit behaviour and for practical design. The class-A power amplifier is a related circuit that operates with the transistor biased in its linear region, enabling it to deliver an undistorted output at the cost of efficiency. This report investigates both configurations as part of a laboratory assignment on BJT common-emitter and class-A power amplifiers. The objectives are to measure AC and DC voltages in a common-emitter amplifier, to determine voltage gain, input impedance, and output impedance, and to calculate and measure input and output power for a class-A amplifier. The analysis combines theoretical calculations with experimental measurements in order to compare expected and observed performance, and to highlight the effects of practical limitations.

II. Part 1-4

A. Common Emitter DC Bias

The first step of the analysis was to measure the resistor values used in the circuit. The results are summarised in Table I.

TABLE I
Measured resistor values

Parameter	Value	Unit
R_{E}	0.9981	kΩ
R_{C}	3.0008	kΩ
R_2	9.9911	kΩ
R_1	33.423	kΩ

Using these resistance values, the approximate operating point of the transistor can be calculated. Since the base current I_{B} is much smaller than the current through R_2, the base voltage is approximated as shown in (1).

$$V_{\mathrm{B}} \approx V_{\mathrm{CC}} \cdot \frac{R_2}{R_1 + R_2} \tag{1}$$

The emitter voltage is then given by (2), assuming a base-emitter drop of $0.7\,\mathrm{V}$.

$$V_{\mathrm{E}} = V_{\mathrm{B}} - V_{\mathrm{BE}} \tag{2}$$

The emitter current is determined from Ohm's law according to (3).

$$I_{\mathrm{E}} = \frac{V_{\mathrm{E}}}{R_{\mathrm{E}}} \tag{3}$$

As $I_{\mathrm{C}} \gg I_{\mathrm{B}}$, the collector current is approximated as equal to the emitter current. The collector voltage is therefore given by (4).

$$V_{\mathrm{C}} = V_{\mathrm{CC}} - (I_{\mathrm{C}} \cdot R_{\mathrm{C}}) \tag{4}$$

Finally, the intrinsic emitter resistance r_{e} is calculated using (5), which is provided in the assignment as Eq. 3.1.

$$r_{\mathrm{e}} = \frac{25\,\mathrm{mV}}{I_{\mathrm{E}}} \tag{5}$$

The results of these calculations are listed in Table II.

TABLE II
Calculated node voltages and emitter current

Parameter	Value	Unit
V_{CC}	10.000	V
V_{B}	2.287	V
V_{E}	1.587	V
V_{C}	5.228	V
I_{E}	1.590	mA
r_{e}	15.722	Ω

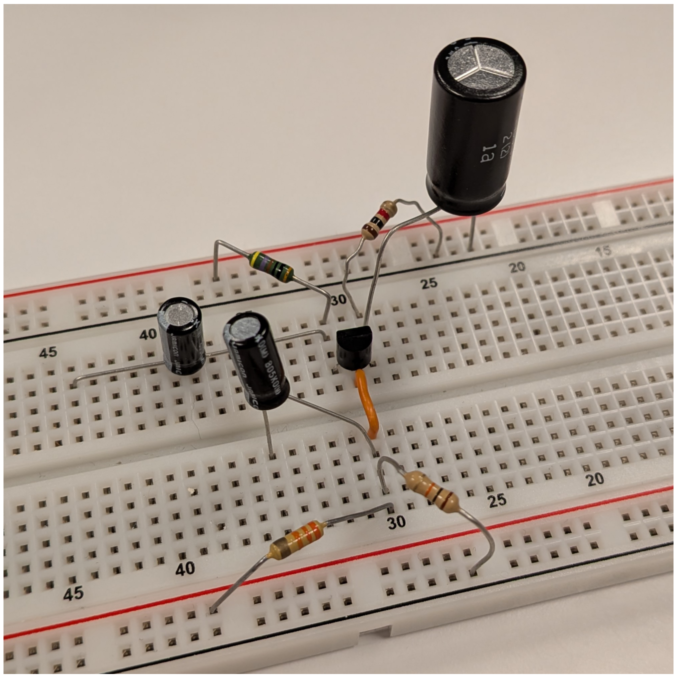

Fig. 1. Photograph of the constructed Voltage-Divider Bias amplifier circuit.

The constructed circuit is shown in Figure 1. The measured node voltages were recorded, and the corresponding values of I_{E} and r_{e} were calculated, as shown in Table III.

TABLE III
MEASURED NODE VOLTAGES AND EMITTER CURRENT

Parameter	Value	Unit
V_{CC}	9.98	V
V_{B}	2.215	V
V_{E}	1.541	V
I_{E}	1.544	mA
V_{C}	5.38	V
r_{e}	16.192	Ω

To compare measured and theoretical results, the relative difference is calculated using (6). In this expression, M denotes the measured value and C denotes the calculated (theoretical) value. The difference is expressed as a percentage of the calculated value.

$$\text{Difference } (\%) = \frac{M - C}{C} \cdot 100 \tag{6}$$

For this part of the experiment, only the intrinsic emitter resistance r_{e} was compared. The measured value of r_{e} differed from the calculated value by $2.99\,\%$.

B. Voltage Gain

The voltage gain of the amplifier can be estimated theoretically using the intrinsic emitter resistance. According to the assignment, the expression for A_{V} is given in (7), which is referred to as Eq. 3.2.

$$A_{\mathrm{V}} = -\frac{R_{\mathrm{C}}}{r_{\mathrm{e}}} \tag{7}$$

The input signal specified in the assignment is $V_{\mathrm{sig}} = 20\,\mathrm{mV}$. However, this value resulted in a distorted output, so the signal generator was set to $14\,\mathrm{mV}$ to avoid distortion.

The output amplitude V_{o} was measured and the experimental voltage gain was calculated using (8). This expression follows directly from the definition of voltage gain as the ratio of output to input. The negative sign indicates the $180°$ phase inversion of the signal, which also allows direct comparison with the theoretical gain.

$$A_{\mathrm{V}} = -\frac{V_{\mathrm{o}}}{V_{\mathrm{sig}}} \tag{8}$$

The calculated and measured values are summarised in Table IV. The relative difference in A_{V} was calculated using Eq. 6.

TABLE IV
CALCULATED AND MEASURED VOLTAGE GAIN

Parameter	Value	Unit
V_{sig}	13.7	mV
V_{o}	2.14	V
A_{V} (calc.)	−185.321	—
A_{V} (meas.)	−156.204	—
Difference	−15.71	%

C. Input Impedance

The input impedance of the amplifier can be estimated theoretically using (9), which is referred to as Eq. 3.3 in the assignment. The data sheet specifies a transistor current gain in the range 150–400. An approximate value of $\beta = 250$ was chosen for this calculation.

$$Z_{\mathrm{i}} = R_1 \parallel R_2 \parallel (\beta \cdot r_{\mathrm{e}}) \tag{9}$$

The input impedance was also determined experimentally by inserting a series resistor of $1\,\mathrm{k\Omega}$ at the input and setting the signal generator to $20\,\mathrm{mV}$. From the resulting division of voltage between Z_{i} and R_{x}, the input impedance is obtained according to (10).

$$Z_{\mathrm{i}} = \frac{V_{\mathrm{i}}}{V_{\mathrm{sig}} - V_{\mathrm{i}}} \cdot R_{\mathrm{x}} \tag{10}$$

The results of the theoretical and experimental calculations are summarised in Table V. The relative difference in Z_{i} was calculated using Eq. 6.

TABLE V
CALCULATED AND MEASURED INPUT IMPEDANCE

Parameter	Value	Unit
R_{x}	0.9863	kΩ
V_{sig}	19.7	mV
V_{i}	14.0	mV
Z_{i} (calc.)	2.6466	kΩ
Z_{i} (meas.)	2.4225	kΩ
Difference	−8.47	%

D. Output Impedance

The output impedance is first estimated theoretically using (11), which corresponds to Eq. 3.4 in the assignment.

$$Z_{\mathrm{o}} = R_{\mathrm{C}} \tag{11}$$

The experimental value of Z_{o} was obtained by loading the output with a resistor and using the voltage-divider relationship between Z_{o} and R_{L}. Following the same approach as for Z_{i}, the expression in (12) is derived from the divider that produces the loaded output V_{L} from the unloaded V_{o}.

$$Z_{\mathrm{o}} = \frac{V_{\mathrm{o}} - V_{\mathrm{L}}}{V_{\mathrm{L}}} \cdot R_{\mathrm{L}} \tag{12}$$

The signal generator was set to $14\,\mathrm{mV}$ to avoid distortion, with a load of $R_{\mathrm{L}} = 3\,\mathrm{k\Omega}$ connected. The measured quantities and the resulting impedances are summarised in Table VI. The relative difference in Z_{o} was calculated using Eq. 6.

TABLE VI
CALCULATED AND MEASURED OUTPUT IMPEDANCE

Parameter	Value	Unit
R_{L}	3.001	kΩ
V_{sig}	13.7	mV
V_{o}	2.14	V
V_{L}	1.08	V
Z_{o} (calc.)	3.0008	kΩ
Z_{o} (meas.)	2.9454	kΩ
Difference	−1.85	%

III. POWER AND EFFICIENCY ANALYSIS

A. DC Bias Analysis

The circuit is rewired to use a different transistor and resistor values. The measured resistor values are listed in Table VII.

TABLE VII
MEASURED RESISTOR VALUES FOR PARTS 5–6

Parameter	Value	Unit
R_{E}	20.07	Ω
R_{C}	134.02	Ω
R_2	176.2	Ω
R_1	986.6	Ω

The bias point was estimated using the same relations introduced earlier, namely Eqs. 1–4. The resulting calculated values are summarised in Table VIII.

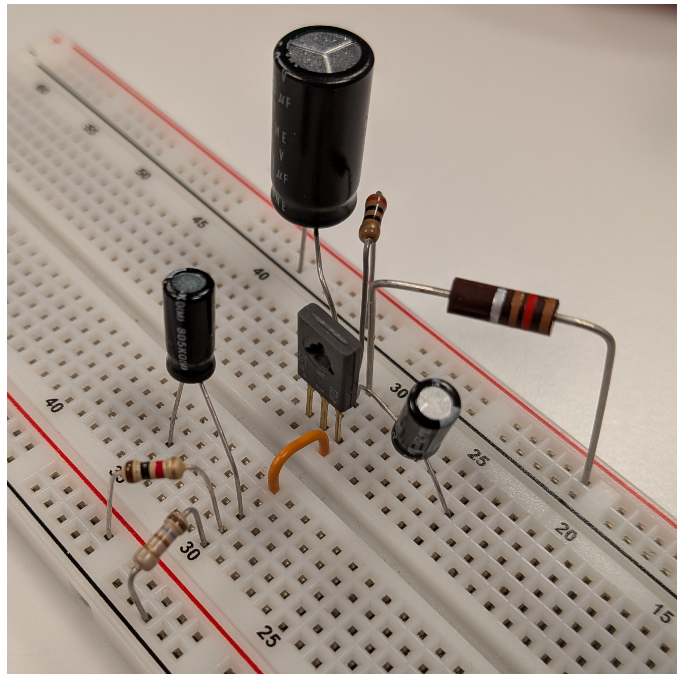

Fig. 2. Photograph of the constructed Class-A amplifier circuit.

TABLE VIII
CALCULATED DC BIAS VALUES

Parameter	Value	Unit
V_{CC}	10.000	V
V_{B}	1.515	V
V_{E}	0.815	V
V_{C}	4.556	V
I_{E}	40.623	mA

The circuit was then constructed using the measured resistors. A photograph of the assembled circuit is shown in Figure 2. From this circuit, the node voltages were measured and the emitter current was derived. The results are presented in Table IX.

TABLE IX
MEASURED DC BIAS VALUES

Parameter	Value	Unit
V_{CC}	9.97	V
V_{B}	1.426	V
V_{E}	0.811	V
V_{C}	4.60	V
I_{E}	40.409	mA

B. Power and Efficiency

The DC input power is obtained from (13), which is Eq. 3.5 in the assignment. Since only the emitter current I_{E} was determined, it is used here as an approximation for I_{C}.

$$P_{\mathrm{i}}\ (\mathrm{DC}) = V_{\mathrm{CC}} \cdot I_{\mathrm{C}} \tag{13}$$

The maximum theoretical output swing depends on the supply voltage V_{CC} and the DC bias point at the collector V_{C} and is given by (14). The expression gives V_{o} as voltage peak-to-peak.

$$V_{\mathrm{o}} = \begin{cases} 2 \cdot V_{\mathrm{C}}, & \text{if } V_{\mathrm{C}} < \frac{1}{2} \cdot V_{\mathrm{CC}} \\ 2 \cdot (V_{\mathrm{CC}} - V_{\mathrm{C}}), & \text{if } V_{\mathrm{C}} > \frac{1}{2} \cdot V_{\mathrm{CC}} \end{cases} \tag{14}$$

As V_{C} is less than half the supply voltage, the upper equation is used. Due to some small losses in the circuit the found voltage is also rounded down to whole volts.

The maximum AC output power is then estimated using (15), which is the peak-to-peak form of Eq. 3.6 in the assignment. This form is used since the calculated V_{o} represents the maximum peak-to-peak output voltage swing.

$$P_{\mathrm{o}}\ (\mathrm{AC}) = \frac{V_{\mathrm{o}}^2\ (\text{p-p})}{8 \cdot R_{\mathrm{C}}} \tag{15}$$

The efficiency is then calculated using (16), which is Eq. 3.7 in the assignment.

$$\eta = \frac{P_{\mathrm{o}}}{P_{\mathrm{i}}} \cdot 100 \tag{16}$$

The calculated values are listed in Table X.

TABLE X
CALCULATED POWER AND EFFICIENCY

Parameter	Value	Unit
V_{o}	9.0	V
P_{i}	406.23	mW
P_{o}	75.548	mW
η	18.60	%

For the experimental measurements, the input was increased until the maximum undistorted output was achieved, corresponding to the 40 mV rms setting of the signal generator. The resulting input and output voltages were measured, both as rms values as well. The collector resistor voltage drop is calculated as in (17), and the collector current from Ohm's law as in (18).

$$V_{\mathrm{RC}} = V_{\mathrm{CC}} - V_{\mathrm{C}} \tag{17}$$

$$I_{\mathrm{RC}} = \frac{V_{\mathrm{RC}}}{R_{\mathrm{C}}} \tag{18}$$

The measured input power is again calculated from Eq. 13, and the output power from (19), which this time uses the rms form of Eq. 3.6.

$$P_{\mathrm{o}}\ (\mathrm{AC}) = \frac{V_{\mathrm{o}}^2}{R_{\mathrm{C}}} \tag{19}$$

The efficiency is then obtained from Eq. 16. All measured values are summarised in Table XI.

TABLE XI
MEASURED POWER AND EFFICIENCY

Parameter	Value	Unit
V_{i}	23.0	mV
V_{o}	2.38	V
P_{i}	399.48	mW
P_{o}	42.27	mW
η	10.58	%

Finally, the percentage differences between calculated and measured values of P_{i}, P_{o}, and η are given in Table XII, calculated using Eq. 6.

TABLE XII
RELATIVE DIFFERENCES BETWEEN CALCULATED AND MEASURED POWER AND EFFICIENCY

Parameter	Difference %	Unit
P_{i}	−1.66	%
P_{o}	−44.06	%
η	−43.11	%

IV. CONCLUSION

The investigation of BJT common-emitter and class-A power amplifiers showed that theoretical analysis can predict bias voltages, gain, and impedance with good accuracy. Most calculated values matched measurements within about ten per cent, confirming the reliability of the basic circuit equations. For the class-A amplifier, the measured efficiency was significantly lower than the calculated maximum, mainly due to non-ideal device behaviour and practical measurement limits. These results highlight the difference between ideal calculations and real performance, and show the importance of experimental verification in amplifier design.